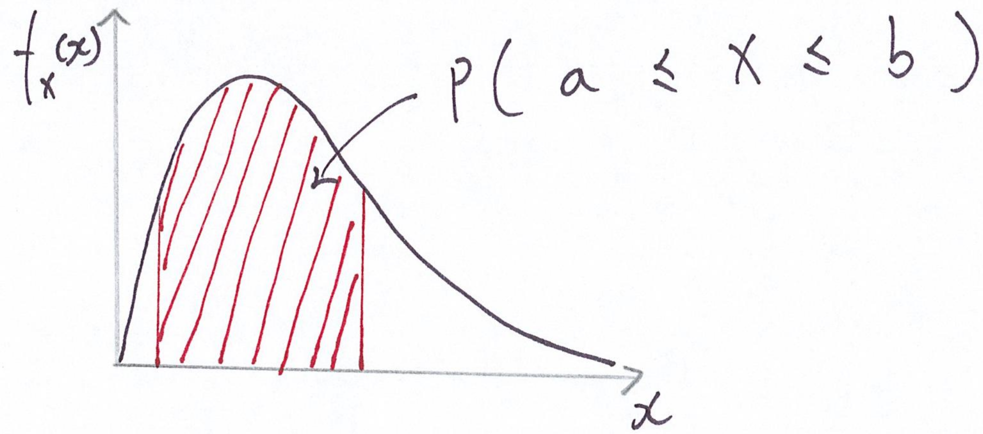
Continuous random variables

Def. The probability density function (PDF) of a con't RV is such that

$$P(a \le X \le b) = \int_a^b f_X(x)\, dx$$



Properties of PDF's:

- $P(X=c) = P(c \le X \le c) = \int_c^c f_X(x)\, dx = 0$

☆ PMF's can not be used to study con't RV's.

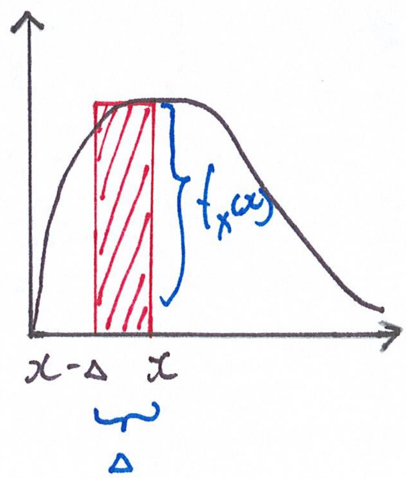
- $f_X(x) \geq 0$ and $f_X(x)$ can be greater than 1.

- $\int_{-\infty}^{\infty} f_X(x)\, dx = 1$

How to interpret a PDF?

- Let $\Delta > 0$ be a small # (e.g. $\Delta = 10^{-15}$)

- $P(X \in [x-\Delta, x])$
  $\simeq \Delta \cdot f_X(x)$



$\Rightarrow f_X(x) = \dfrac{P(X \in [x-\Delta, x])}{\Delta}$

= the prob per unit length.

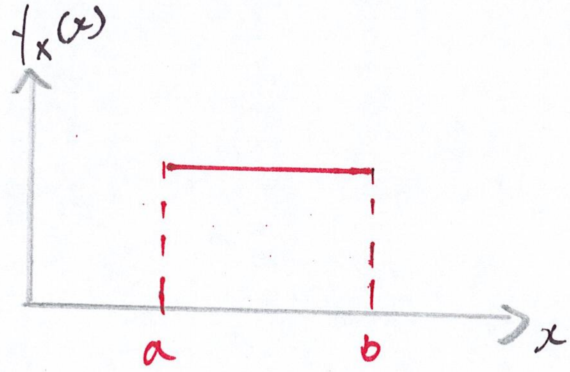
Probability $\Updownarrow$ Distance

Δ $\Updownarrow$ Time

PDF $\Updownarrow$ Speed

Ex. Suppose that $f_X(x)$ is flat over $[a, b]$.

Find $F_X(x)$.



$$\because \int_{-\infty}^{\infty} f_X(x)\, dx = \int_a^b c \cdot dx = 1$$

$$\therefore f_X(x) = c = \frac{1}{b-a}, \quad x \in (a, b).$$

$P(X \leq x)$

$$\therefore F_X(x) = \begin{cases} 0, & x \leq a \\ \int_a^x \frac{1}{b-a}\, dx = \frac{x-a}{b-a}, & x \in (a, b) \\ 1, & x \geq b \end{cases}$$

Ex. Let

$$f_X(x) = \begin{cases} 6x \cdot (1-x), & 0 < x < 1 \\ 0, & \text{o.w.} \end{cases}$$

Find $P(|X - \frac{1}{2}| > \frac{1}{4})$.

$\because$ $|x - \frac{1}{2}| > \frac{1}{4}$

$\Leftrightarrow$ $x - \frac{1}{2} < -\frac{1}{4}$ or $x - \frac{1}{2} > \frac{1}{4}$

$\Leftrightarrow$ $x < \frac{1}{4}$ or $x > \frac{3}{4}$

$\therefore$ $P(|X - \frac{1}{2}| > \frac{1}{4})$

disjoint

$= P(\{X < \frac{1}{4}\} \cup \{X > \frac{3}{4}\})$

$= \underset{①}{P(X < \frac{1}{4})} + \underset{②}{P(X > \frac{3}{4})}$

① $P(X < \frac{1}{4}) = F_X(\frac{1}{4})$

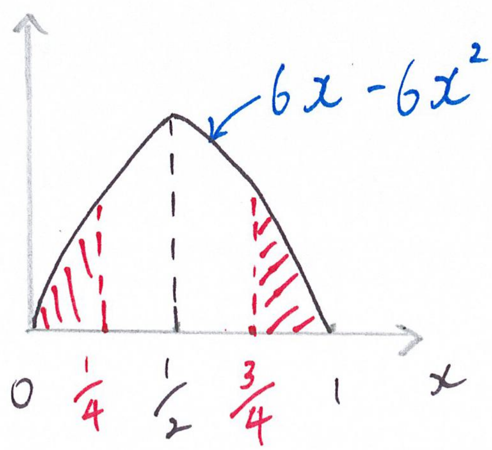
$$= \int_{-\infty}^{\frac{1}{4}} f_X(x)\, dx$$

$$= \int_{0}^{\frac{1}{4}} 6x(1-x)\, dx$$

$$= 6\left(\int_0^{\frac{1}{4}} x\, dx - \int_0^{\frac{1}{4}} x^2\, dx\right)$$

$$= 6 \cdot \left(\frac{1}{2}x^2\Big|_0^{\frac{1}{4}} - \frac{1}{3}x^3\Big|_0^{\frac{1}{4}}\right)$$

$$= 0.15625$$



② $P(X > \frac{3}{4}) = P(X < \frac{1}{4})$

$\therefore P(|X - \frac{1}{2}| > \frac{1}{4}) = 2 \times 0.15625 = 0.3125$